

Departure and destination aerodromes in the OPDI flight list

How they are derived, how accurate they are, and how to use them

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i Who this is for

You are reading this because you use, or are deciding whether to use, the ADEP and ADES columns of the [OPDI flight list](#). It covers what those columns mean, how the pipeline derives them from raw ADS-B, how often they are right, where they are systematically wrong, and what a downstream analysis should do about it.

Every figure on this page is produced by running the pipeline over a sample of ADS-B and scoring the result against EUROCONTROL Network Manager movement data. Nothing is estimated or carried over from an earlier version.

Summary

OPDI reconstructs flights from ADS-B state vectors alone: no flight plans, no schedules, no radar. From the resulting trajectory it must infer where each flight departed from and where it arrived — which is straightforward when the aircraft is seen on the runway at both ends, and hard in every other case.

Three findings shape what the flight list now contains.

Departures and arrivals want different algorithms. A departing aircraft’s first ADS-B fix is usually on or near the runway, so geometry alone settles it. An arriving aircraft’s last fix is often where reception was lost rather than where it landed, so geometry alone does not. The flight list therefore derives departures with one method and arrivals with another.

Candidate aerodromes are chosen by exact distance. The published algorithm narrowed its candidates using a hexagonal spatial index and measured exact distance only among the survivors, so an aerodrome slightly further out was discarded before its distance was ever computed. Measuring first and choosing afterwards is the single change that makes everything else in this study work.

A null is now three different answers. ADEP and ADES can be an aerodrome, the marker OOA for a flight that began or ended outside the observed area, or null for “could not tell”. Companion columns say which. Earlier releases collapsed the last two into a single null, and about one flight in twelve genuinely falls into the second.

1 Reading this page

Section 2 says what the flight list contains. Section 3 documents the pipeline that produces it, step by step. Section 4 defines the measures. Everything after that is evidence.

Terms are defined where they first appear and not before. Two conventions run throughout:

- **legacy** means the algorithm and constants behind every OPDI flight list published before 2026 — reachable in code as `DetectionConfig.legacy()`, so released data stays reproducible.
- **shipped** means what this study recommends and what the pipeline now produces by default.

“Production” is used only to talk about the running system, never as the name of an algorithm.

2 What the flight list contains

One row per flight, where a *flight* is a continuous stretch of ADS-B reception from one aircraft under one callsign. The columns this page is about:

Column	Meaning
ID	The track identifier. Stable within a release, not stable across algorithm versions — see Section 15.
ICAO24	The aircraft’s 24-bit transponder address. The join key to most other aviation data.
FLT_ID	The broadcast callsign. Space-padded to eight characters in the raw feed; trim before joining.
DOF	Date of flight, taken from the first reception.
ADEP / ADES	Departure and destination aerodrome, ICAO code — or OOA, or null.
ADEP_SOURCE / ADES_SOURCE	aerodrome, out_of_area, or undetermined.
ADEP_P / ADES_P	Other aerodromes that were plausible. Comma-separated, ranked behind the chosen one.
FIRST_SEEN / LAST_SEEN	Reception bounds. Not off-block and in-block times.
VERSION	Which algorithm produced the row.

2.1 The three states of an aerodrome column

This distinction matters more than anything else on this page for a downstream analysis, and it is new:

aerodrome An ICAO code. The pipeline saw enough to name it.

out_of_area ADEP/ADES is OOA. The flight entered or left the observed area rather than departing or landing inside it — the trajectory begins or ends at the edge of coverage, not near any aerodrome. **This is an answer, not a failure.** An analysis counting movements should exclude these; one measuring coverage should count them as correct.

undetermined ADEP/ADES is null. The pipeline could not tell. Filter these out; do not impute them.

Collapsing the last two — as every release before v4.0.0 did — makes traffic appear to be missing where in fact it was correctly identified as originating elsewhere.

3 The pipeline that produces it

Everything below is real ADS-B, processed by the code in the `opdi` repository. No step is a re-implementation for the purposes of this report.

The same pipeline, step by step. Each row names the module and the table it writes; every step reads the table written above it unless stated otherwise.

Step	Module	Writes	Does
—	—	<i>OpenSky archive</i>	1 Hz global state vectors
00	<code>reference/</code>	<code>h3_airport_detection_zones</code>	H3 zones — aerodrome zones, OurAirports reference
01	<code>ingestion/osn_statevector_state</code>	<code>osn_statevector_v2</code>	European bounding box, thinning to 5 s
02	<code>pipeline/tracks.py</code>	<code>osn_tracks</code>	Split into tracks, attach the H3 index
02a	<code>cleaning/cleaner.py</code>	<code>osn_tracks_clean</code>	Mask implausible values, keep the rows
03	<code>pipeline/flights.py</code>	<code>opdi_flight_list</code>	ADEP / ADES detection — reads 02a and 00
04	<code>pipeline/events.py</code>	<code>events, measurements</code>	Milestones along each flight
05	<code>output/</code>	<code>published dataset</code>	Parquet and CSV export

3.1 00 — Reference data

Builds what the detection needs to know about aerodromes. The important product is the **H3 detection zone table**: for every aerodrome, the set of hexagonal cells (resolution 7, about 1.5 NM across) covering concentric rings out to a generous radius, each tagged with the ring’s inner and outer distance.

This is a spatial index, not an answer. Its job is to turn “which aerodromes might this fix be near?” into a hash lookup. Distances are measured afterwards, exactly — Section 3.5.1 explains why that distinction turned out to matter more than any threshold in the study.

Aerodrome positions, elevations and scheduled-service flags come from OurAirports. The candidate set is restricted to large and medium aerodromes; Section 16 quantifies what that excludes.

3.2 01 — State-vector ingestion

Reads the OpenSky archive and keeps what OPDI needs:

- **A bounding box**, roughly 25.9° W to 49.7° E and 26.7° N to 70.3° N. This is the *observed area*, and its edge is what the 00A marker refers to.
- **Thinning to 5 s**. The archive is a complete 1 Hz grid. OPDI keeps the newest row in each 5 s bin — a **bin-based** rule. Earlier releases used a fixed-phase rule, keeping a row only if one existed at the exact second divisible by five. Section 11 measures the difference.

The raw global 1 Hz feed is never stored.

3.3 02 — Track building

Groups state vectors into flights and assigns `track_id`. Two aircraft transmissions belong to the same track when they share an aircraft address and callsign and are not separated by a gap of more than 30 minutes, or more than 15 minutes below 5,000 ft — the second catching an aircraft that keeps its transponder on while parked.

This rule is frozen. Changing it changes `track_id` for all published data, so it is not tuned and not swept.

The step also attaches each fix’s H3 cell at resolution 7, which is what makes the aerodrome join in step 03 a lookup rather than a scan.

3.4 02a — Trajectory cleaning

Masks implausible values to null and keeps the row: duplicate state vectors, out-of-range positions, stale broadcasts (ADS-B sends position and velocity in separate messages, so identical consecutive values mean *repeated*, not *measured*), derivative spikes, and isolated points. It also marks segment boundaries at coverage gaps so nothing downstream interpolates across them.

Masking rather than deleting is deliberate: it preserves the distinction between “no data” and “interpolated data”. It does mean the detection has fewer usable samples, since a fix with no barometric altitude cannot vote on a trend. Section 10 measures what that costs and what it buys.

3.5 03 — Flight list

Where ADEP and ADES are decided. Two algorithms, one per role.

3.5.1 What both share

Candidate selection. Every aerodrome whose detection zone the aircraft passed through is a candidate. The exact great-circle distance from the trajectory’s closest approach to each candidate is computed, and the smallest wins. **No candidate is eliminated before its distance is measured.**

That sentence is the study’s central finding, and it was not true before. The legacy algorithm kept only the candidates in the innermost occupied hexagon ring, then measured exact distance among those. A ring steps about 2.8 NM, so an aerodrome a kilometre further out was routinely

discarded unmeasured — and the higher the aircraft, the more aerodromes share a ring and the less a ring index can separate them. Section 5 shows what that was worth.

The scheduled-service penalty. An aerodrome without scheduled commercial service has a fixed distance added to its effective distance before that comparison, so a military or general-aviation field must be *clearly* nearest rather than merely nearest. It re-ranks candidates without changing how many flights are answered, so it can move accuracy and cannot move coverage.

3.5.2 endpoint — for departures

Take the aerodrome nearest the track’s first fix, and accept it only if that fix is within a radius and either on the ground or below a height *above field elevation* — above the aerodrome’s own elevation, since a fixed altitude cut-off means nothing at a field sitting at 5,000 ft. Failing that, if the fix sits at the edge of the observed area, answer **00A**; otherwise abstain.

Its abstention is **geometric**: it declines when the endpoint is not where a departure’s endpoint should be.

3.5.3 trend — for arrivals

For every fix below a flight-level cap and within a radius of a candidate aerodrome, ask whether smoothed barometric altitude is rising or falling. Each fix votes. If descending votes beat climbing votes by more than a margin, that aerodrome is a destination.

Its abstention is **evidential**: it declines when the altitude trace near an aerodrome does not clearly rise or fall. That is why it suits arrivals — it uses the whole approach rather than one fix, and one fix is exactly what an arrival is least likely to have in the right place.

Two refinements complete it. Among candidates whose distances differ by less than a small band, the one the aircraft’s course actually points at wins (Section 12). And where **trend** cannot name an aerodrome, the same border test **endpoint** uses can still distinguish “came from outside” from “cannot tell”.

3.6 04–05 — Events and export

The flight list becomes the spine for milestone extraction — take-off, landing, top of climb, level segments, flight-level crossings — and is exported as parquet and CSV. Neither step changes ADEP or ADES.

4 How the numbers are measured

Two stages, and the distinction governs how much any figure is worth.

The research sweep. Hundreds of parameter combinations applied to a cached intermediate table. Cheap enough to be exhaustive, which is what makes a grid of this size possible — but it re-implements the decision rule rather than running it, so it can only *propose* a value.

The pipeline run. The proposed values run through the same code path that writes the published flight list, and scored identically. Nothing in this report becomes a default without passing this stage.

4.1 The sample

Three days, **5–7 June 2025**, with **5–7 June 2024** as an independent second period. Both are full-day samples inside the bounding box, thinned to 5 s.

Ground truth is EUROCONTROL Network Manager movement data for the same days, matched to OPDI tracks on aircraft address plus callsign plus date, and where several tracks match, the one starting closest to the reference off-block time. It is not redistributable, which is where reproducibility stops (Section 17).

4.2 The measures

For each role — departures and arrivals — separately:

Coverage Of the reference flights, the share OPDI gives *any* answer for. OOA counts as an answer.

Accuracy Of the answers given, the share that are right.

Correct / wrong The same two quantities as counts, which is what makes settings comparable. Coverage and accuracy always move in opposite directions, and neither ratio alone says whether an exchange was worth taking.

Score $\text{correct} - k \times \text{wrong}$, with $k = 2$. A judgement, stated so it can be disagreed with: a wrong aerodrome is worse than a silence because it corrupts *two* movement counts — the one it invents and the one it omits — and is invisible downstream, where a silence corrupts one and is an explicit null anyone can filter on.

A change is therefore worth taking exactly when it wins more than k correct answers per wrong one it introduces.

Out-of-area recall and precision are reported separately, because a method could otherwise buy headline accuracy by labelling OOA liberally. Recall is the share of genuinely out-of-area flights labelled as such; precision is the share of OOA labels that are correct. Where the report says *per-aerodrome recall* later on, that is a different measure and is defined where it appears.

The 2025 sample carries **95,116** reference flights and the 2024 sample **92,799**.

5 What each change is worth

Each row adds one change to the row above **and keeps it**, so the marginal worth of a change is the difference between adjacent rows, and the last row is the shipped pipeline exactly. That is asserted before the run rather than checked afterwards: the harness refuses a ladder whose final rung is not `DetectionConfig()` in its thresholds, its per-role algorithms and its track table, and refuses one containing a rung that changes nothing.

Read the *signs* across the two periods before the sizes. A change that helps on one sample and hurts on the other has not been demonstrated, whatever its magnitude on the sample it was tuned against.

5.1 2025

	Step	Departures	Δ dep	Arrivals	Δ arr	Total
legacy algorithm, uncleaned tracks		63,064	—	60,603	—	123,667
+ rank candidates by exact distance		62,539	−525	60,255	−348	122,794
+ cut the radius by distance, not hexagon band		62,512	−27	60,255	0	122,767
+ smooth altitude before the flight-level cut		62,714	+202	60,211	−44	122,925
+ scheduled-service penalty		63,563	+849	61,072	+861	124,635
+ tuned flight-level cap		67,185	+3,622	61,748	+676	128,933
+ tuned vote margin		67,533	+348	61,980	+232	129,513
+ tuned radius		67,727	+194	62,000	+20	129,727
+ bearing tie-break		67,910	+183	63,092	+1,092	131,002
+ out-of-area labelling		69,072	+1,162	62,265	−827	131,337
+ departures from endpoint		71,806	+2,734	61,486	−779	133,292
+ tuned endpoint radius		72,013	+207	61,486	0	133,499
+ cleaned tracks (= shipped)		72,013	0	61,009	−477	133,022

Scores, not ratios: coverage and accuracy always move in opposite directions, and only the counts say whether an exchange was worth taking. Net over the whole ladder: **+9,355**.

Rows one to ten produce both roles with **trend**, so each change is attributable to its own line. Row eleven switches departures to **endpoint**, which is why the departure column moves there and barely moves before it — the trend parameters above are tuned for arrivals.

5.2 2024

	Step	Departures	Δ dep	Arrivals	Δ arr	Total
legacy algorithm, uncleaned tracks		53,569	—	48,411	—	101,980
+ rank candidates by exact distance		53,335	−234	48,183	−228	101,518
+ cut the radius by distance, not hexagon band		53,326	−9	48,178	−5	101,504
+ smooth altitude before the flight-level cut		53,597	+271	48,200	+22	101,797
+ scheduled-service penalty		53,765	+168	51,482	+3,282	105,247
+ tuned flight-level cap		57,115	+3,350	53,178	+1,696	110,293
+ tuned vote margin		57,393	+278	53,414	+236	110,807
+ tuned radius		57,561	+168	53,167	−247	110,728
+ bearing tie-break		58,068	+507	53,701	+534	111,769
+ out-of-area labelling		58,634	+566	53,456	−245	112,090
+ departures from endpoint		62,980	+4,346	52,742	−714	115,722
+ tuned endpoint radius		63,046	+66	52,742	0	115,788
+ cleaned tracks (= shipped)		63,040	−6	52,156	−586	115,196

Scores, not ratios: coverage and accuracy always move in opposite directions, and only the counts say whether an exchange was worth taking. Net over the whole ladder: **+13,216**.

Rows one to ten produce both roles with **trend**, so each change is attributable to its own line. Row eleven switches departures to **endpoint**, which is why the departure column moves there and barely moves before it — the trend parameters above are tuned for arrivals.

6 The parameter sweeps

Both periods run on identical grids, and cells are ranked by the two together — each period’s score divided by its own reference count so the larger sample cannot dominate. A single-period argmax appears only as a stability check, because over hundreds of cells the top cell moves under noise even when the surface underneath is stable.

Departures — best cells on both periods together

Rank	Flight-level cap	Vote margin	Radius (NM)	2025 score	2024 score
1	FL125	0	30	70,249	59,562
2	FL125	0	20	70,295	59,488
3	FL100	0	20	69,978	59,643
4	FL125	0	40	70,026	59,365
5	FL125	2	20	69,995	59,374

Arrivals — best cells on both periods together

Rank	Flight-level cap	Vote margin	Radius (NM)	2025 score	2024 score
1	FL60	0	30	61,183	52,752
2	FL75	0	30	60,947	52,943
3	FL60	0	40	61,124	52,713
4	FL75	0	40	60,860	52,909
5	FL60	0	60	61,013	52,588

The endpoint rule, which is what departures use

Role	2025 best	2024 best	Agrees
Departures	30 NM / 15,000 ft	30 NM / 15,000 ft	yes
Arrivals	30 NM / 10,000 ft	30 NM / 10,000 ft	yes

6.1 The flight-level cap, followed to the top

Flight-level cap	2025 departures	2025 arrivals	2024 departures	2024 arrivals
FL25	56,929	58,891	48,543	48,413
FL40	65,101	60,788	55,101	51,161
FL50	67,301	61,453	57,220	52,429
FL60	68,279	61,793	58,183	52,893
FL75	69,230	61,858	58,945	53,389
FL100	70,142	60,248	59,696	51,886
FL125	70,334	58,314	59,579	50,327
FL150	69,756	56,861	58,752	48,892
FL175	68,768	55,184	57,839	47,042
FL200	67,603	53,185	56,643	44,841
FL225	66,148	51,414	55,157	42,580
FL250	64,361	49,492	53,547	40,099

Flight-level cap	2025 departures	2025 arrivals	2024 departures	2024 arrivals
FL275	62,738	47,834	51,977	38,041
FL300	60,742	45,510	50,052	35,653

At 20 NM, through the pipeline itself. Departures peak at FL125 and arrivals at FL75 on the 2025 sample. Coverage keeps rising with the cap — a higher cap admits more fixes — but accuracy falls faster, because fixes taken higher up are further from whichever aerodrome turns out to be the answer.

7 What ships, and how well it works

Configuration	Departures by	Arrivals by	Dep. cover- age	Dep. accu- racy	Arr. cover- age	Arr. accu- racy	Total
SHIPPED — endpoint departures, trend arrivals	endpoint	trend	79.96%	98.23%	67.57%	98.31%	133,022
endpoint for both roles	endpoint	endpoint	80.03%	98.14%	70.15%	96.22%	131,032
trend for both roles	trend	trend	74.10%	98.96%	68.72%	98.18%	130,072
legacy — the algorithm published before 2026	trend	trend	67.46%	98.82%	66.99%	98.19%	122,158
nearest aerodrome, unconditionally	nearest	nearest	91.64%	87.68%	88.55%	79.29%	86,837

Each row is a complete configuration — which algorithm names departures, which names arrivals — not a step in a sequence. The rows are alternatives to choose between.

All five read the **cleaned** tracks, so this table compares algorithms on one input. That is why the **legacy** row here does not match the ladder’s first rung in Section 5: this one is the legacy *algorithm* on current input, and that one is the legacy algorithm on the input it actually shipped with. The pair separates the algorithm from the data, and neither number alone does.

The shipped configuration scores **133,022 against legacy’s 122,158**, a gain of +10,864.

7.1 Which changes hold on both samples

A change measured on one sample has been observed; a change measured on two has been *demonstrated*. The ladder runs identically on both periods, so every rung can be graded on whether its sign survives.

Step	2025 Δ total	2024 Δ total	Verdict
+ rank candidates by exact distance	−873	−462	consistently costs no material effect
+ cut the radius by distance, not hexagon band	−27	−14	
+ smooth altitude before the flight-level cut	+158	+293	holds
+ scheduled-service penalty	+1,710	+3,450	holds

Step	2025 Δ total	2024 Δ total	Verdict
+ tuned flight-level cap	+4,298	+5,046	holds
+ tuned vote margin	+580	+514	holds
+ tuned radius	+214	-79	disagrees
+ bearing tie-break	+1,275	+1,041	holds
+ out-of-area labelling	+335	+321	holds
+ departures from endpoint	+1,955	+3,632	holds
+ tuned endpoint radius	+207	+66	holds
+ cleaned tracks (= shipped)	-477	-592	consistently costs

Δ is the change in **total** score, departures plus arrivals, against the rung above. A rung moving less than 48 flights on both samples is reported as having no material effect rather than having its sign read off noise.

+ **tuned radius** changes sign between the two samples. A value that gains on the sample it was chosen against and loses on an independent one has not been demonstrated, and this study treats that as a reason to revisit it rather than to average the two.

Consistently costs is not the same as *wrong*, and the two rungs carrying that verdict are worth reading rather than skipping.

Exact-distance ranking costs on its own. It is not an improvement; it is an *enabler*. Two rungs later the flight-level cap is worth over four thousand on both samples, and under the hexagon-index selection it replaces, the same change lost ground. The cost here buys that, and quoting this rung alone tells the story backwards.

Cleaning costs too, and the report does not hide it. Masking implausible values removes candidate fixes, so coverage falls. What it buys is accuracy and a trustworthy input for every other thing built on tracks. Whether that exchange is worth it at $k = 2$ is a judgement, and the number is here so a reader can disagree with it — see Section 10.

Out-of-area labelling reads as *holds*, which corrects an easy misreading: it does cost arrival score, and it gains more on departures than it costs on arrivals, because flights beginning outside the observed area are now named correctly instead of being silently dropped. Section 9 gives both halves.

7.2 Does it hold on the second period?

Configuration	2025 total	2024 total
legacy — the algorithm published before 2026	122,158	100,591
SHIPPED — endpoint departures, trend arrivals	133,022	115,193

No parameter was chosen against one period and checked on the other: both were swept on identical grids and ranked together.

8 What should change next

Three of the values this report describes are **not** what the evidence supports. They are stated here rather than quietly altered: a default should not be changed by the document that measures it, and a reader deserves to know which parts of what ships are settled and which are not.

Change	Evidence	Confidence
Arrival detection radius 20 NM → 30 NM	pipeline, both samples (+214 and −79); sweep ranked jointly prefers 30 NM	high — four independent measurements agree
Out-of-area on trend → off, or stricter	arrival precision 50.35% against endpoint ’s 89.20%; costs arrivals −827 and −245	high — a label right half the time is worse than a null
Vote margin 2 → 0	research sweep only, both samples	medium — never tested through the pipeline

The radius. It gains on the sample it was chosen against and loses on an independent one, and the research sweep ranked over both periods prefers 30 NM as an interior optimum. This is the single rung in Section 5 whose sign does not survive a second sample.

Out-of-area on arrivals. The label is correct about half the time, against 89% for the same test applied to departures. The asymmetry is geometric: a track that *begins* at the edge of the observed area almost certainly entered from outside, while one that *ends* near it may equally be a flight still bound somewhere inside whose reception was lost. Every arrival label replaces a silence, so at $k = 2$ a coin flip is a losing trade. Either the arrival test gains a direction requirement — the course is already computed for the bearing tie-break — or arrival out-of-area is left to **endpoint**.

The vote margin. The sweep prefers 0 to the shipped 2 on both samples, and the pipeline has never tested that step: the ladder walks $4 \rightarrow 2$ and stops. Cleaning is the likely reason the margin has stopped earning its keep — it exists to stop noise flipping a climb-or-descent call, and cleaning now removes much of that noise before the vote is counted. Worth one pipeline run before adopting.

9 Out-of-area, and the three states

Role	Truly out-of-area	Labelled out-of-area	Recall	Precision	In-area accuracy
Departures	8.30%	0.74%	7.96%	89.20%	98.58%
Arrivals	8.37%	0.44%	2.67%	50.35%	98.33%

In-area accuracy is accuracy restricted to flights whose true aerodrome is a real one — detection skill with the free out-of-area wins removed, and the number to read when comparing naming rules.

10 What cleaning costs and buys

Cleaning is the last rung of the ladder rather than a separate experiment, so what it is worth is measured *given everything else already adopted* — which is the only form of the question that bears on a decision. The two rows differ in nothing but the track table.

2025

Role	Raw coverage	Clean coverage	Raw accuracy	Clean accuracy	Δ score
Departures	79.96%	79.96%	98.23%	98.23%	0
Arrivals	68.39%	67.57%	98.17%	98.31%	−477

2024

Role	Raw coverage	Clean coverage	Raw accuracy	Clean accuracy	Δ score
Departures	72.69%	72.69%	97.82%	97.82%	−6
Arrivals	62.48%	61.61%	96.99%	97.07%	−586

Cleaning **masks implausible values to null and keeps the row**, and the detection path drops fixes with no barometric altitude — so it can only remove candidate samples. Coverage falling and accuracy rising is the expected shape; whether that exchange is worth taking at $k = 2$ is what the score column says.

11 The 5 s sampler

The bin-based rule is adopted because it is the **correct** rule, not because it measures better. It does not depend on the OpenSky archive being a complete 1 Hz grid, which is a property of the archive rather than a guarantee; the fixed-phase rule it replaces survives only because that property currently holds, and degrades badly if it ever stops.

i This figure is not re-measured here, and why

Version 6 compared the two samplers across every parameter cell of two complete runs — 371 cells of the trend grid and 126 of the endpoint grid, identical grids over identically-treated input — and found the largest median coverage difference anywhere to be a few *thousandths* of a percentage point, against a decision bar of 0.5.

This report does not repeat that measurement, and says so rather than restating the number as if it had. There is no fixed-phase arm left to compare against — the bin-based rule is now what the pipeline does — and these sweeps run on cleaned tracks over a different flight-level grid, so a comparison against version 6's arm would be confounded by the cleaning and the grid rather than measuring the sampler. Two arms on different data is exactly the error that produced version 6's worst figure, and manufacturing a comparison here would repeat it.

Where the sampler *should* matter is not here at all. Aerodrome detection reads an altitude trend over tens of samples, and one extra fix per five-second bin cannot change the sign of a trend. Milestones defined by a single instant on the ground — off-block, take-off, landing, in-block — are a different case, where a recovered bin is the difference between having a fix and not. That measurement belongs to the events study and has not been made.

One consequence is not reversible by configuration: **track_id values differ** between the two samplers, because rescued rows sit at track boundaries and the splitter breaks on gaps.

12 The bearing tie-break

This is the newest component in the algorithm and the best evidenced. It was proposed by the research sweep, declined once for want of a pipeline measurement, and is adopted here because `process_dai` confirms it on both samples.

Among candidates whose distances differ by less than a small band — that is, already equidistant to within the measurement — the tie is broken by *alignment*: which aerodrome the aircraft's course actually points at. Outside the band, distance decides alone.

The band is the whole mechanism. Alignment cannot *name* an aerodrome, because every field on the same radial behind the right one is equally well aligned; it can only separate two that distance has already declared equal.

13 Using the columns

Filter on the source columns, not on nulls. `ADES_SOURCE == 'aerodrome'` is the set of named arrivals; `'out_of_area'` is the set correctly identified as leaving the observed area; `'undetermined'` is the set to drop.

Expect under-count, not over-count. Abstention is the dominant failure mode, so an aerodrome's movement count from OPDI is a lower bound. Section 14 gives the ratio per aerodrome for the hundred busiest.

Join on ICA024 plus trimmed FLT_ID plus date. Callsigns are space-padded to eight characters in the raw feed.

FIRST_SEEN is not off-block. It is when reception began.

Do not compare ID across versions. See Section 15.

14 By aerodrome, for the busiest hundred

Being regenerated.

The matching is flight by flight, not count against count. **Per-aerodrome recall** is, of the flights whose *true* aerodrome is X, the share OPDI attributes to X. **Count ratio** is what OPDI assigns to X divided by X's true movements — it includes flights that truly belong elsewhere, so it is what an operational count would show, and it can exceed one.

Role	Movements in truth	Recall, shipped	Recall, legacy	Median count ratio
Departures	65,382	91.52%	82.05%	0.987
Arrivals	65,360	78.94%	79.11%	0.899

15 Versions, and what is reproducible

The `VERSION` column says which algorithm produced a row.

Version	What it means
v2.0.0	Legacy trend , both roles.
v3.0.0	Legacy endpoint . In a merged list these two were coalesced into one column, so a row's version told you about the departure side only.
v4.0.0	This study: exact-distance ranking, per-role algorithms, tuned thresholds, the bearing tie-break, out-of-area labelling, and cleaned tracks as input. One string for the whole flight list.

`DetectionConfig.legacy()` still returns every pre-2026 value, so re-processing a month reproduces the aerodromes a release published.

 **track_id** is not reproducible across the sampler change

The 5 s thinning rule changed at the same time. Rescued rows sit at track boundaries and the splitter breaks on gaps, so a track that gains a fix can be split differently. Re-ingesting an old month reproduces the *aerodromes* but not the *identifiers*. Do not join ID across releases.

16 Limitations

What is known to be missing or fragile, stated rather than implied.

- **One shipped value is contradicted by this study's own evidence.** The arrival detection radius ships at 20 NM. Measured through the pipeline it gains 20 flights on the 2025 sample — noise on 95,116 — and loses 247 on 2024, and the research sweep ranked over both periods prefers 30 NM as an interior optimum. 20 NM was a single-period argmax and it did not hold. Section 7 states the recommendation; the default has not been changed inside this report, because a value should not be altered by the document that measures it.
- **The vote margin is the least-supported shipped value.** The sweep prefers 0 over the shipped 2 on both periods, and the pipeline has never tested that step — it walks 4 → 2 and stops. Cleaning is the likely reason: the margin exists to stop noise flipping a climb/descent call, and cleaning now removes much of that noise before the vote is counted.
- **Out-of-area labelling costs arrival score, and the cost is a measurement artefact.** Adding the label adds rows to the flight list, and the benchmark pairs each reference flight with the track starting closest to its off-block time — so a new row can take a pairing from a track that carried a real answer. Detection did not get worse. This is reported rather than corrected away, because a benchmark that quietly adjusts its own alignment to flatter a result is worth less than one that shows where it is fragile.

- **The cleaning comparison is one-sided.** Cleaning is measured as the ladder’s last rung, and departures show no change there — `endpoint` reads a candidate cache built from cleaned tracks in every rung, so the earlier rungs were never really “raw” for that role. The figure is the cost to arrivals, not to the pipeline as a whole.
- **The legacy algorithm is not exactly reproducible.** Two identical runs of it named three arrivals differently out of 95,116. Ring-count selection leaves candidates tied on an integer, and the tie is resolved by an ordering that does not separate them. Exact-distance ranking removes those ties rather than arbitrating them, and every configuration using it reproduced exactly.
- **The candidate set is large and medium aerodromes.** A flight whose true aerodrome is a small field or a heliport cannot be named by any method here, whatever its thresholds, because the aerodrome is not a candidate. These flights are counted in every denominator on this page, so each coverage figure already carries their cost.
- **The border margin is fixed at 30 NM.** It decides what counts as leaving the observed area. Changing it redefines the ground truth as well as the method, so a sweep would move the target along with the aim.
- **H3 resolution 7 is fixed.** It no longer decides which candidate wins — that is exact distance now — but it still decides which aerodromes become candidates.
- **Two three-day samples.** Both in June. Seasonal effects are not measured.
- **Ground truth covers about ninety aerodromes.** Per-aerodrome figures are meaningful only for those.

17 Provenance

Every CSV behind this page carries the script, arguments, commit and source-file fingerprint that produced it, in `data/_manifest.json`. An output with no entry is reported as unverified rather than shown as fact.

17.1 Where reproducibility stops

Everything on this page can be recomputed from the OpenSky archive and the code in the `opdi` repository, with one exception: the EUROCONTROL Network Manager movement data used as ground truth is not redistributable. It is an external input, and every accuracy figure here is measured against it.